

Trying CoMOLA from Mac Terminal:

- 1) Go to the terminal and install Homebrew: `/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`
(Copy pasting this, might put the code in two different line in the terminal. Make sure it's in one line.)
- 2) Install python 3.11 using: `brew install python@3.11`
- 3) Usually mac has git pre installed. Check via: `git --version`
If not available, use: `brew install git`
- 4) Check for R: `R --version` → Should give something like: `/usr/local/bin/R` (preferred location). If not available, use: `brew install r` → Should give something like: `/opt/homebrew/bin/R`
- 5) Navigate to the location where you want to clone the repository:
`cd ~/Documents` # or wherever you want
- 6) Clone the repo via:
`git clone --branch lecture --single-branch https://git.ufz.de/agriscape/CoMOLA.git`
- 7) Navigate within the CoMOLA folder with: `cd CoMOLA`
- 8) Create python environment: `python3.11 -m venv comola`
- 9) Activate the environment: `source comola/bin/activate`
→ you should now see: (comola) yourname@MacBook CoMOLA %
- 10) Install following python dependencies:
`pip install --upgrade pip`
`pip install numpy==1.26.4 matplotlib`
- 11) Now ask for R and python paths via:
`which R`
`which python`
- 12) Out of terminal: Go to file explorer and within CoMOLA folder open `config.ini` file. You might need a code editor in case you do not have one already. You can use Visual Studio Code: <https://code.visualstudio.com/download> and then open the config.ini file via the editor.
- 13) Once the `config.ini` file is open, Change R and Python path to the ones above from step 11 and save the file. You can also use the editor to change parameters for experimentation accordingly.
- 14) Back in terminal, you should now be able to run the optimization via: `python __init__.py`

NOTE: For future use, you can simply activate the environment via step 9 (source comola/bin/activate), then go to the location of CoMOLA folder via Step 5 and 7 (in our case: `cd ~/Documents` and then `cd CoMOLA`) and then finally running the optimization via step 14 (`python __init__.py`).

In case step 9 returns that no environment with the name comola exists when you restart your terminal in future, that might mean that mac is not saving the environment. In this situation, you need to repeat step 5, 7, 8, 9 and 10 and then run optimization via step 14.

Other errors to look out for:

- 1) In case you are getting a similar error saying **.DS_Store** directory does not exist.

```
CoMOLA — -zsh — 119x40
...op/LSS/CoMOLA — -zsh  ...laenv/bin/jupyter-lab - R  .../LSS/CoMOLA — -zsh  .../LSS/CoMOLA — -zsh  +

10-02-2026 13:59:27 | 3, [2, 6, 6, 1, 1, 2, 2, 3, 4, 4, 4, 6, 6, 1, 5, 5, 3, 6, 7, 4, 2, 1, 1, 4, 4, 3, 3, 5, 5, 5, 2,
6, 6, 1, 1, 1, 1, 5, 7, 7, 2, 5, 4, 3, 3, 1, 7, 7, 4, 6, 4, 3, 1, 5, 2, 2, 2, 6, 7, 5, 3, 1, 5, 5, 6, 6, 3, 4, 7, 5, 1,
1, 2, 2, 6, 7, 3, 4, 7, 3, 3, 3, 2, 6, 7, 7, 3, 2, 2, 1, 1, 1]

10-02-2026 13:59:27 | 4, [3, 6, 7, 5, 5, 3, 3, 7, 5, 4, 7, 6, 6, 5, 6, 4, 7, 7, 7, 4, 3, 7, 1, 4, 6, 1, 4, 6, 2, 1, 2,
7, 6, 7, 5, 4, 4, 7, 7, 7, 6, 3, 5, 6, 2, 6, 7, 7, 3, 6, 5, 6, 4, 3, 6, 6, 7, 6, 7, 7, 2, 2, 6, 5, 6, 6, 3, 4, 7, 5, 1,
1, 2, 2, 6, 6, 3, 4, 7, 3, 3, 3, 2, 6, 7, 7, 3, 2, 2, 1, 1, 3]

10-02-2026 13:59:28 | Evaluation of individual 2
10-02-2026 13:59:28 | Evaluation of individual 1

10-02-2026 13:59:28 | Update genome in models folder 1

10-02-2026 13:59:28 | Update genome in models folder 2

10-02-2026 13:59:28 | Error in executing external model /Users/hanno/Desktop/LSS/CoMOLA/models/.DS_Store. Directory doe
s not exist.

10-02-2026 13:59:28 | Error: <class 'SystemError'>, Error in executing external model. Directory does not exist.

10-02-2026 13:59:28 | Error in executing external model /Users/hanno/Desktop/LSS/CoMOLA/models_1/.DS_Store. Directory d
oes not exist.

10-02-2026 13:59:28 | Error: <class 'SystemError'>, Error in executing external model. Directory does not exist.

10-02-2026 13:59:28 | not_accepted_ind []

10-02-2026 13:59:28 | Fitness values are: [[44.0, 61.0436036279331, 90.5242580637318, 10.180840023202], [44.0, 61.04360
36279331, 90.5242580637318, 10.180840023202], [44.0, 61.0436036279331, 90.5242580637318, 10.180840023202], [44.0, 61.04
36036279331, 90.5242580637318, 10.180840023202]]

Best Individual: [2, 6, 6, 1, 1, 2, 2, 3, 4, 4, 4, 6, 6, 1, 5, 5, 3, 6, 7, 4, 2, 1, 1, 4, 4, 3, 3, 5, 5, 5, 2, 6, 6, 1,
1, 1, 1, 5, 7, 7, 2, 5, 4, 3, 3, 1, 7, 7, 4, 6, 4, 3, 1, 5, 2, 2, 6, 7, 5, 3, 1, 5, 5, 6, 6, 3, 4, 7, 5, 1, 1, 2, 2
, 6, 6, 3, 4, 7, 3, 3, 3, 2, 6, 7, 7, 3, 2, 2, 1, 1, 1] : [44.0, 61.0436036279331, 90.5242580637318, 10.180840023202]

10-02-2026 13:59:28 | The generation of a new candidate needed 0 seconds.
```

Go to the models folder within CoMOLA and delete the **.DS_Store** file and run the optimization again.